



The Algerian Olympiad in Artificial Intelligence

Basics Roadmap

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Important Note

This roadmap is intended to serve as a **rough guideline** for what should be learned during preparation for the Algerian Olympiad in Artificial Intelligence. It is not a strict syllabus, nor does it impose a single mandatory order in which topics must be studied.

There is **no single resource** that covers everything needed, and there is **no exact linear path** that must be followed. In practice, participants will often find themselves jumping between topics, temporarily treating certain concepts or tools as black boxes in order to experiment and build intuition, and only later returning to fully understand the underlying theory and inner workings.

Learning Artificial Intelligence is an iterative process, and partial understanding at early stages is both normal and expected.

Participants are also encouraged to actively ask questions and seek help through:

- The official training or community Discord servers.
- Online platforms such as Reddit subreddits related to AI and machine learning.
- **Stack Overflow** for programming-related questions.
- **Math Stack Exchange** for mathematical concepts.

When exploring additional resources independently, participants are strongly advised to regularly consult the **official IOAI website** and its published syllabus. This helps ensure that study time is focused on topics that are actually covered by the competition and avoids spending excessive time on material that is outside the scope.

In particular, care should be taken not to dive too deeply into topics that require mainly **practical familiarity and high-level theoretical understanding**, rather than full mathematical or implementation-level depth.

The IOAI scientific board also provides an official spreadsheet containing recommended educational resources. Participants are encouraged to refer to it when selecting learning materials.

Finally, this roadmap represents **one possible approach** to preparation. Learners are free to use alternative resources or learning paths that they personally find clearer, more comfortable, or more effective, as long as the core concepts and skills are eventually mastered.

Programming Foundations

Programming is the first and most essential pillar of preparation for the Algerian Olympiad in Artificial Intelligence. All participants are expected to develop strong implementation skills in **Python**, with an emphasis on correctness, clarity, and algorithmic thinking.

Python Basics

The initial introduction to Python should be fast, accessible, and practical. For this purpose, the following resource is recommended:

- **W3School Python Tutorial:** Used as a quick and beginner-friendly introduction to Python syntax, data types, control structures, functions, and basic object-oriented concepts.

At this stage, the goal is not mastery, but rather becoming comfortable with reading and writing Python code fluently.

Implementation Practice

To ensure that participants can correctly and efficiently implement algorithms, regular problem-solving practice is required.

- Solve **Codeforces problems rated around 800**.
- Focus on clean implementation, handling edge cases, and writing readable code (at least for yourself).

To maintain implementation consistency and deepen understanding, participants are encouraged to:

- Re-implement common algorithms from scratch from time to time.
- Avoid over-reliance on black-box libraries (this is not the case during competitions).
- When appropriate, use `numpy` to manually implement operations for better numerical intuition.

Numerical Computing and Data Manipulation

Once basic Python proficiency is achieved, participants should move to numerical and data-centric libraries.

NumPy and Pandas

The following resources are recommended:

- **Kaggle Learn** courses on NumPy and Pandas (free and interactive).
- **“Python for Data Analysis”** by Wes McKinney (creator of Pandas).

These resources are sufficient to build a strong foundation in:

- Array-based computing
- Vectorized operations
- DataFrames and structured data handling
- Basic data cleaning and transformation (scaling and such)

Exploratory Data Analysis (EDA)

Exploratory Data Analysis should be practiced continuously alongside learning NumPy and Pandas. Recommended practice methods include:

- Selecting random datasets from **Kaggle** and performing EDA.
- Comparing personal approaches with solutions from competition winners.
- Using datasets from **OpenML** to practice data inspection, visualization, and preprocessing.

The goal is to develop intuition about data, distributions, anomalies, and feature behavior as there isn't that much theory to it you must focus on practicing with real data.

Mathematics

A strong mathematical background is essential for understanding machine learning models, optimization, and reasoning behind algorithms.

Core Mathematics

Participants are expected to thoroughly study the book:

- **“Mathematics for Machine Learning”**

They should:

- Carefully read all chapters.
- Solve as many exercises as possible (no need to spend a lot of time on pure numerical examples).
- Study the second part of the book to understand real-world applications and gain intuition.

Intuition Through Visual Explanations

To complement formal mathematical study, the following video resources are strongly recommended:

- **3Blue1Brown:**
 - Essence of Linear Algebra
 - Essence of Calculus
 - Essence of Probability

These playlists (and other videos from #SoMe4) help build deep intuition behind abstract concepts.

Statistics

For statistical concepts, the StatQuest YouTube Channel is recommended for topics such as:

- Probability distributions
- Expectation and variance
- Bias-variance tradeoff
- Statistical interpretation of machine learning models

Classical Machine Learning

After acquiring sufficient mathematical maturity, participants should move to classical machine learning algorithms.

Theoretical Foundations

Two primary academic courses are recommended:

- **Stanford CS229** recommended for participants with strong mathematical confidence.
- **MIT 6.036** a more guided and accessible alternative.

These courses cover:

- Supervised and unsupervised learning
- Optimization methods
- Probabilistic models
- Model evaluation and generalization

Practical Machine Learning

Practical implementation should be done using:

- Additional **Kaggle Learn** courses on machine learning.
- **Chapter 5 (Machine Learning)** from “Python for Data Analysis”.

At this stage, participants should become comfortable with:

- The `scikit-learn` framework
- Training and evaluating classical ML models
- Proper data preprocessing and validation

Deep Learning and IOAI-Style Tasks

Deep learning forms the core of most modern Artificial Intelligence systems and plays a central role in IOAI-style problems. At this stage, participants are expected to transition from purely classical machine learning to neural network-based models, while simultaneously becoming familiar with the structure and expectations of official Olympiad tasks.

Introduction to Deep Learning

The following academic resources are recommended for building a solid foundation in deep learning and also some of the main stream topics in Computer Vision:

- **MIT 6.S191 Introduction to Deep Learning**
- **Stanford CS231n Convolutional Neural Networks for Visual Recognition**
- “**Deep Learning**” by Ian Goodfellow et al.

Together, these resources cover:

- Neural network fundamentals
- Backpropagation and optimization
- Convolutional neural networks
- Practical deep learning design choices

Participants are not expected to memorize every theoretical detail immediately; instead, the focus should be on understanding core ideas and developing practical intuition.

IOAI-Style Tasks

In parallel with studying deep learning theory, participants should begin working on problems that resemble the actual format of the International Olympiad in Artificial Intelligence.

- Official IOAI-style tasks hosted in a **GitHub repository maintained by the IOAI Scientific Committee**.
- Tasks and problem statements from **first-round or national selection competitions** of other participating countries.

These problems are designed to test:

- Practical model design
- Data understanding and preprocessing
- Experimental reasoning
- Performance analysis and reporting

At this stage, it is acceptable for certain components to be treated as black boxes initially, as long as participants gradually build a deeper understanding of how and why models work through experimentation and analysis.